

# Light IoT, LED & Smart Signage Applications

**Aug. 2026**  
**Jaesang Cha, Chiba Univ.**

# Agenda

1. Light IoT & Smart LED Applications Concepts
2. Optical Camera Communication(OCC)
3. LED & Signage/ Display Applications

Appendix, Additional technologies & applications



# Light IoT & Smart LED Applications Conctsep





# Light IoT: Concept and Background (1/2)

Using light-emitting things around us to connect, communicate, and create smart applications.

## 1. Light is Everywhere

- ✓ There is no place without light.
- ✓ Many man-made light sources exist in our daily life.



Bulb / Lamp



LED Lighting



LED Display



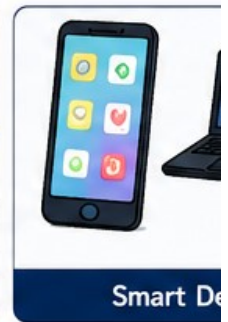
Signage / Ads



Traffic Light



Automotive Light



Smart Device



These light-emitting things are not only for illumination, but transmit data. This enables many useful applications in our daily life.



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## 2. What is Light IoT?

Light IoT (Light-based Internet of Things) is a new paradigm that connects objects and people using light for both sensing and communication.

### Concept of Light IoT

Light Source  
(LED / Display / Lamp)



Data Transmission  
Using Light

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Receiver  
(Camera / Sensor /  
Smart Device)



### Light IoT Enables



Connection of  
Things



Real-time  
Sensing



High-speed  
Communication



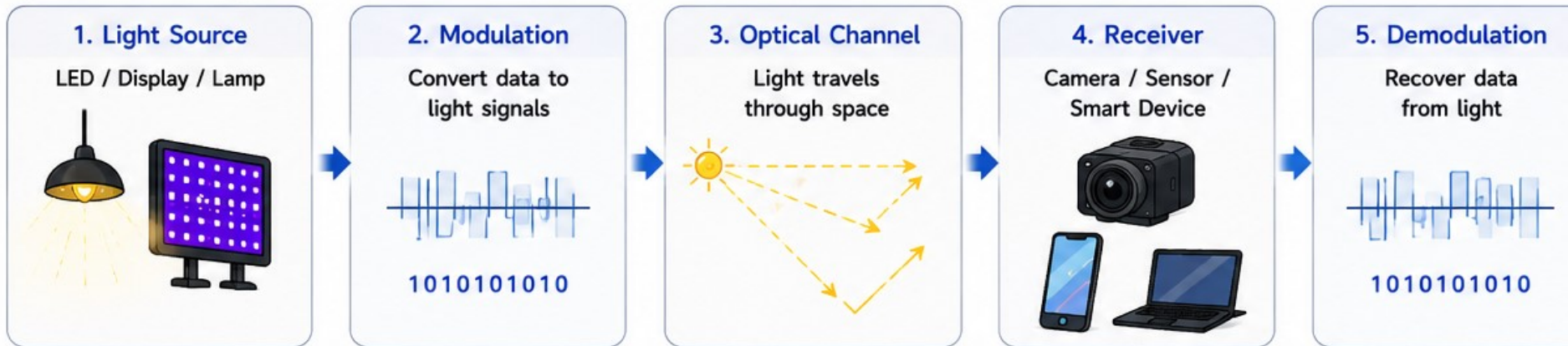


# Light IoT: Concept and Background (2/2)

From light transmission to smart applications in everyday life

Light  
n  
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## 3. How Light IoT Works



## 4. Applications of Light IoT



## 5. Summary

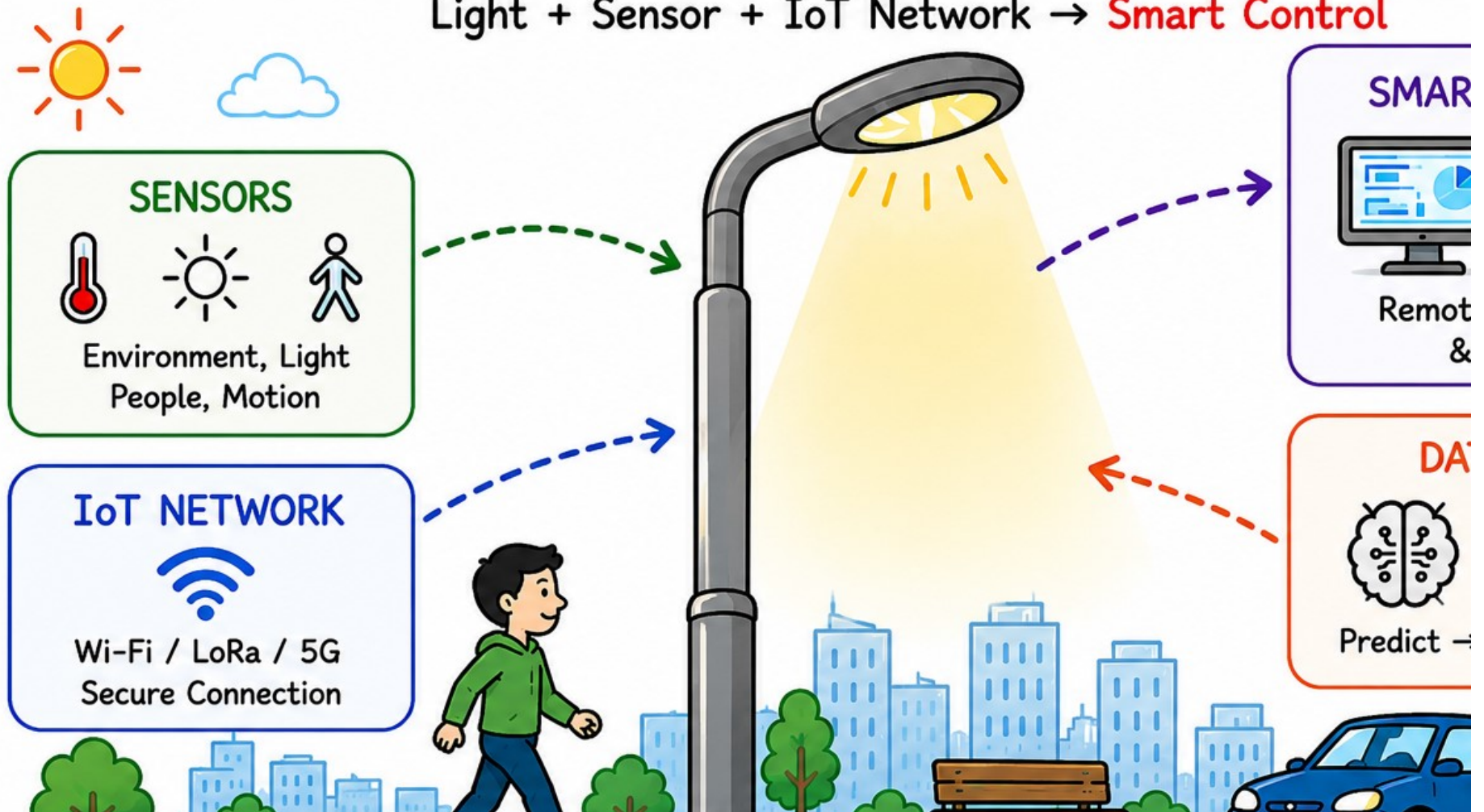


The  
By  
as  
we  
use



# Example of Light IoT Architecture Using Streetlight (Light-based IoT)

Light + Sensor + IoT Network → Smart Control





# LED Technology: Applicable to Various Applications

From simple illumination to smart platforms, LED technology supports many useful services

## Simple Lighting



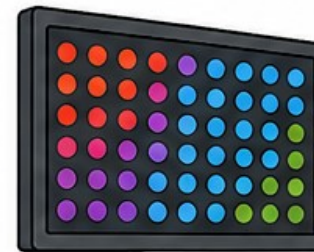
Basic illumination  
for everyday life



## Smart Lighting



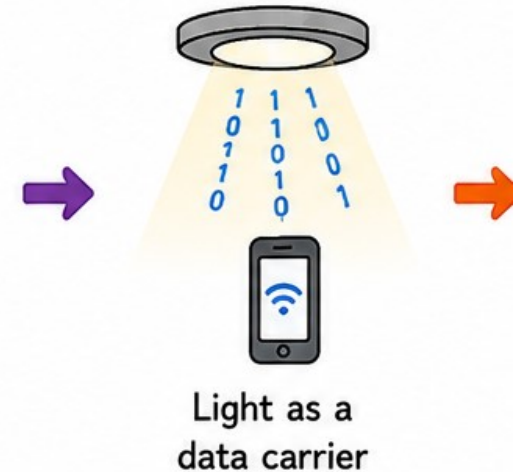
## LED Display



Information  
visualization



## Light Communication



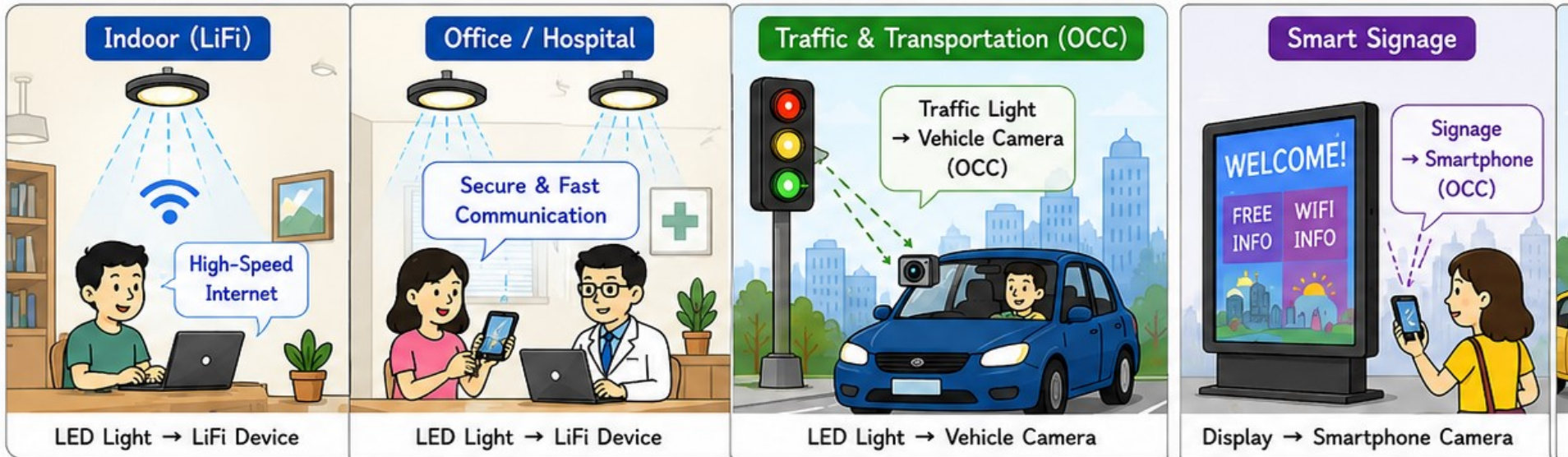
LED technology can be applied to lighting, display, communication, and smart

Lighting → Display → Communication → Smart Signage



# Optical Wireless Applications

Light can carry both **illumination** and **data**!



Using **visible light**, we can communicate **anywhere, anytime**!

## OPTICAL WIRELESS COMMUNICATION (OWC)

In the early 1990s, researchers at Keio University in Japan began studying this technology and called it **VLC (Visible Light Communication)**. Since the 2010s, the term **OWC (Optical Wireless Communication)** has been widely used in international standardization forums such as IEEE 802 and ITU-R.

OWC solutions can be broadly classified into two main approaches:

### 1. High-Speed (LiFi) Solutions



### 2. Camera Communication (OCC) Solutions



## Why Optical



**High Security**  
(Confined to Light Area)



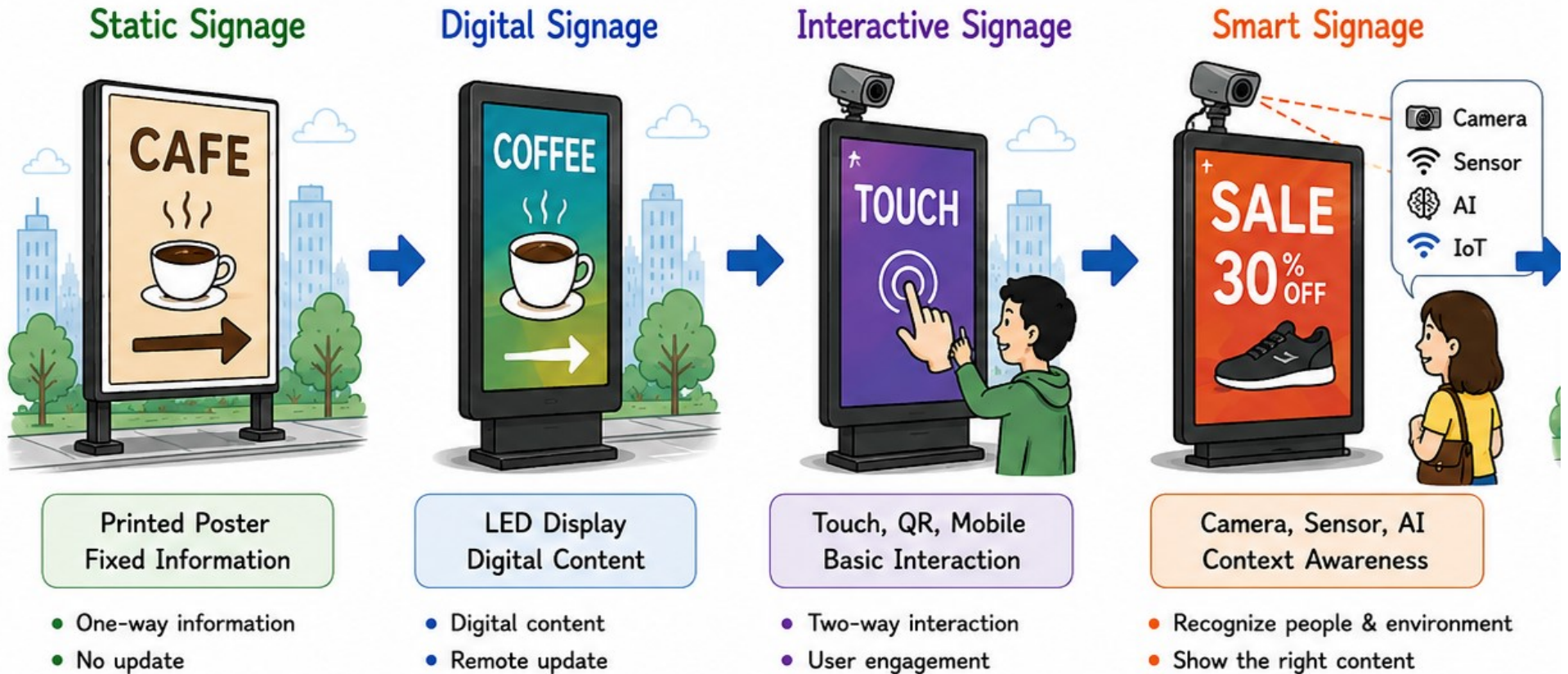
**No RF Interference**  
(Safe for Sensitive Areas)



**Energy Efficient**  
(Low Power LED)

# Various Signage Displays

From static information to intelligent, interactive and immersive displays



Smart signage  
understands





# Various Signage & Display Solutions

The signage and display field is continuously evolving in many forms with the convergence of various ICT technologies.



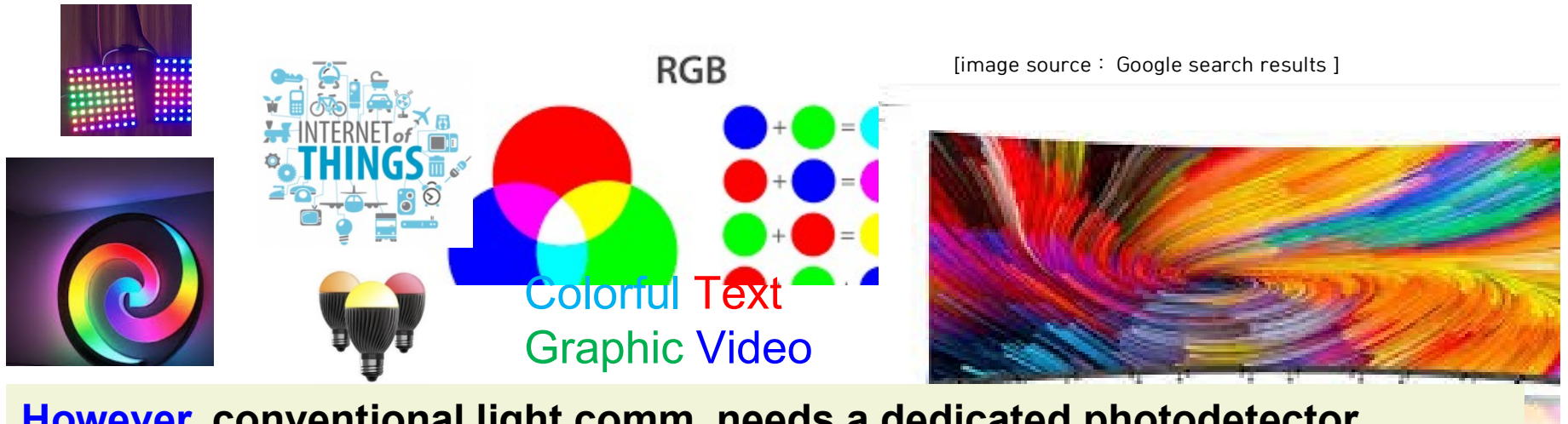
# Optical Camera Communication(OCC)



# Introduction of OCC

*(Technical background)*

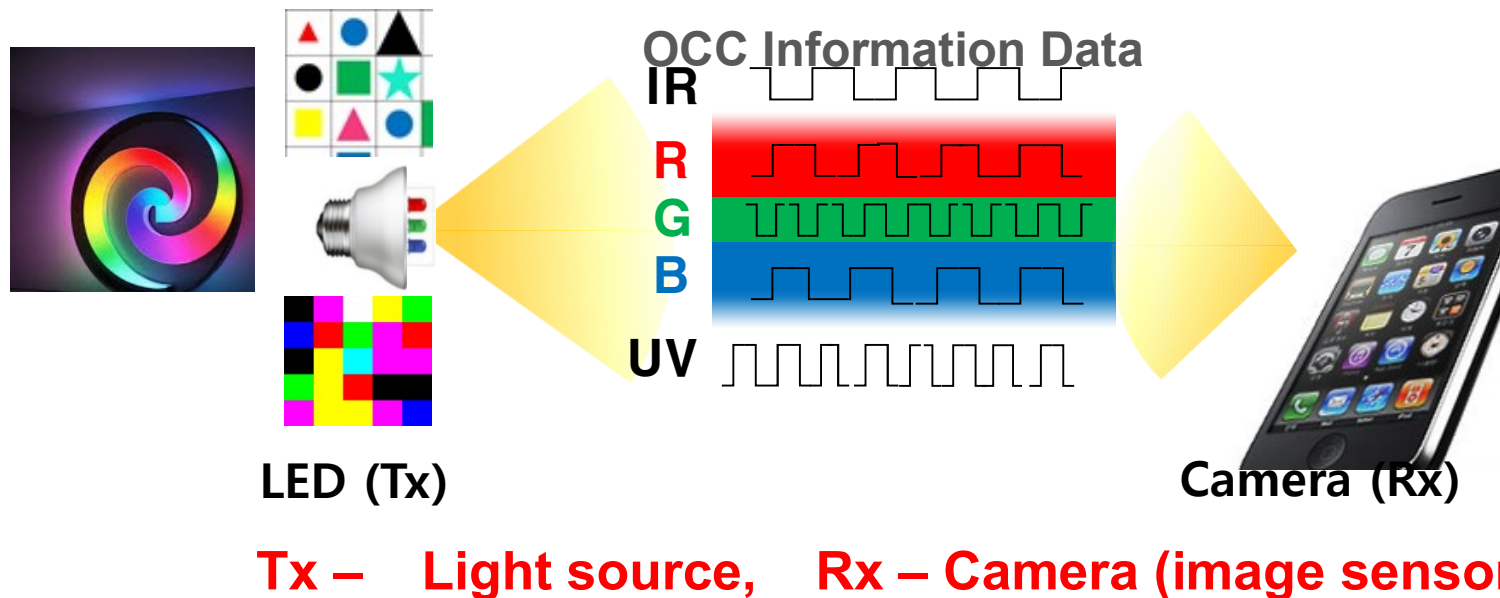
Light-emitting things (e.g., displays) are everywhere. LED lights are bright and easy to see, making them good for data transmission.



**However,** conventional light comm. needs a dedicated photodetector.  
**This is inconvenient** for users.

- Therefore, our research group proposed a camera-based data reception method called Optical Camera Communication (OCC), which was adopted in the IEEE 802.15.7m standard

# Optical Camera Communication (OCC)



- Light Things transmit data and ID information using light
- A smartphone or CCTV camera receives the transmitted light
- Computer Vision extracts the transmitted data and ID information

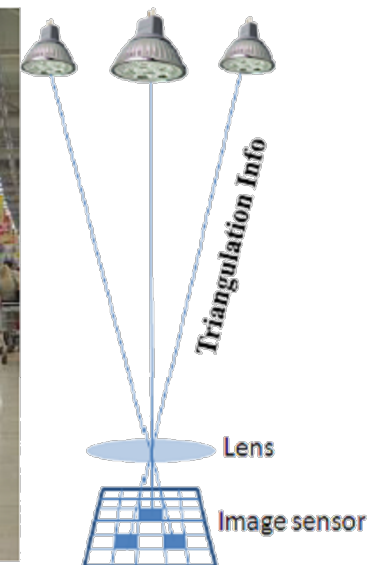


# Concept of OCC Technology

[image source : IEEE802.15.7 intel]



< Line of Sight Marketing >

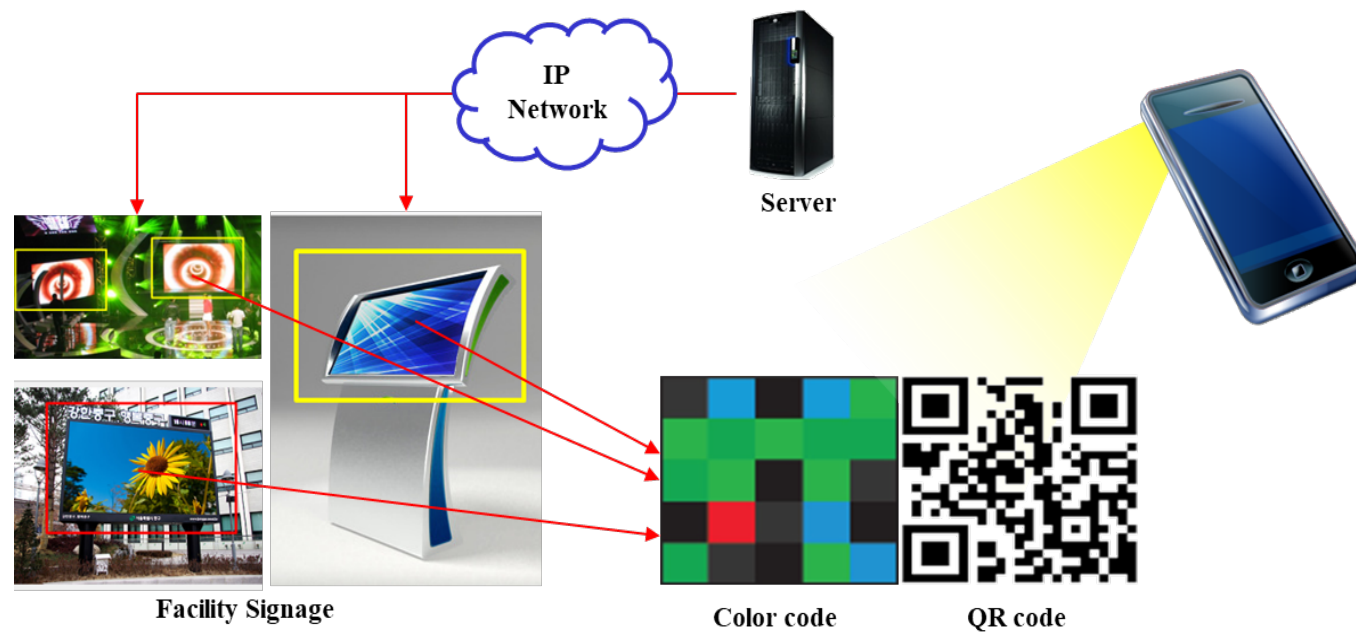


< Indoor Location >

**Tx – Light source, Rx – Camera**

- OCC Technology Provides a low-cost and enhanced user experience

# Use Case : LED Color Tag ID



**Tx : LED color tag ID Rx : Smart Device Camera**



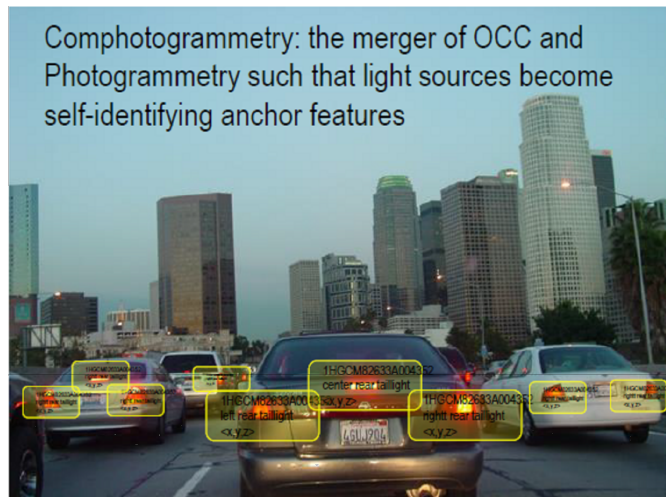
# Use Case : Display Panel Applications



**Tx : Display Panel Rx : Smart Device Camera**

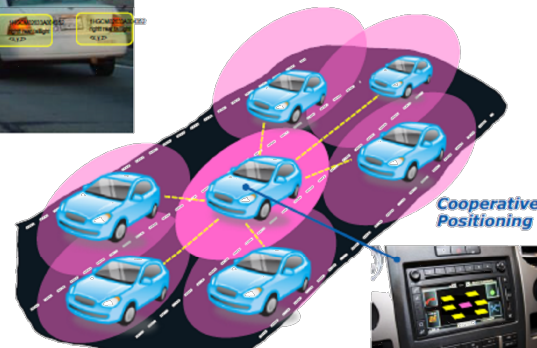
# Use Case : Vehicle-Based Applications

Comphotogrammetry: the merger of OCC and Photogrammetry such that light sources become self-identifying anchor features



[image source : IEEE802.15.7 intel]

High Resolution  
Automobile Positioning



- Various OCC applications using vehicle lights & blackbox cameras are also available

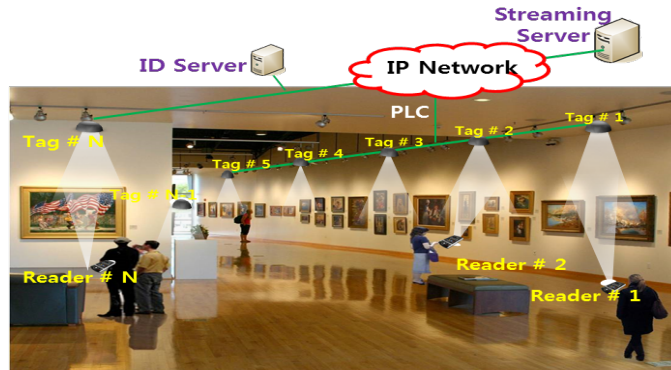
# Use Case : Various OCC Applications #1

[image source : IEEE802.15.7 intel]





# Use Case : Various OCC Applications #2



Exhibition Service



Indoor Store Service



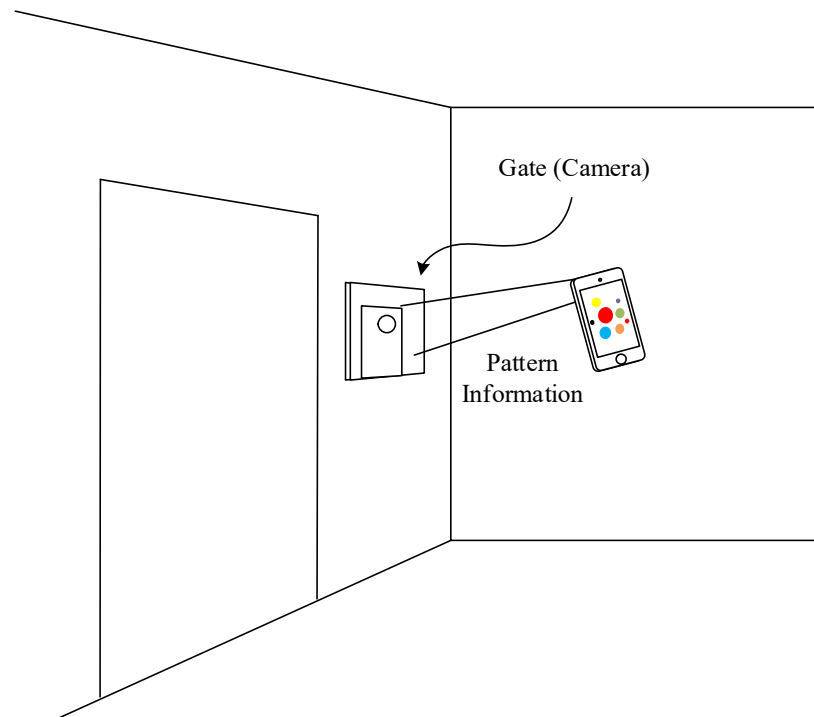
In-flight Service



Facility Signage/Display

- As shown above, OCC can be used in many places for various services

# Use Case : Camera-Based Authentication



- Gate Camera & Smart Device pattern based Authentication application also available